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Exercise 1: SQL Fundamentals

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1.
```
SELECT *
FROM employees;
```

id	first_name	last_name	department	Salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angel
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Fran
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angel
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San Fran

2.
```
SELECT DISTINCT department
FROM employees;
```

department
IT
HR
Finance
Marketing

3.
```
SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;
```

first_name	last_name	salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

4
```
SELECT first_name, last_name, department, salary
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

first_name	last_name	department	salary
Emily	Davis	IT	62000
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000
John	Doe	IT	55000
Sarah	Brown	IT	53000

5.
```
SELECT first_name, last_name, department
FROM employees
WHERE department = 'IT';
```

first_name	last_name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

#1

6. SELECT first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance'
AND salary > 58000;

first_name	last_name	department	salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000

7. SELECT first_name, last_name, department
FROM employees
WHERE department = 'HR'
OR department = 'Marketing';

first_name	last_name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clarke	Marketing

8. SELECT first_name, last_name, department
FROM employees
WHERE department <> 'IT';

first_name	last_name	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clarke	Marketing

9.
```sql
SELECT first_name, last_name, department
FROM employees
WHERE department IN ('HR', 'IT', 'Finance');
```

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

10.
```sql
SELECT first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT'
    AND salary > 50 000
    AND city = 'New York';
```

first_name	last_name	department	salary	city
John	Doe	IT	55 000	New York
Sarah	Brown	IT	63 000	New York

11.
```sql
SELECT first_name, last_name
FROM employees
WHERE department IN ('Finance', 'Marketing')
    AND salary > 52000
ORDER BY salary DESC;
```

first_name	last_name
Mike	Johnson
Robert	Wilson
Daniel	Clark

12. SELECT DISTINCT ~~city~~ city
FROM employees
WHERE department IN ('Finance', 'Marketing');

city
Los Angeles
San Francisco
Houston
Chicago

13. SELECT first_name, last_name, department,
salary, hire_date
FROM employees
WHERE department <> 'Finance'
AND salary > 50000
ORDER BY hire_date ASC;

first_name	last_name	department	salary	hire_date
Emily ~~David~~	Davis ~~White~~	IT ~~Marketing~~	62000 ~~52000~~	2015-02-14 ~~2016-04-10~~
David	White	Marketing	52000	2016-04-10
Jessica	Moore	HR	51000	2018-05-22
John	Doe	IT	55000	2018-06-15
Sarah	Brown	IT	53000	2021-03-28
Daniel	Clark	Marketing	53000	2022-06-01

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14
```
SELECT first_name, last_name, department, city
FROM employees
WHERE (city = 'Chicago' OR city = 'Los Angeles')
    AND department IN ('IT', 'Marketing')
LIMIT 3;
```

first_name	last_name	department	city
Emily	Davis	IT	chicago
Daniel	Clark	Marketing	Chicago